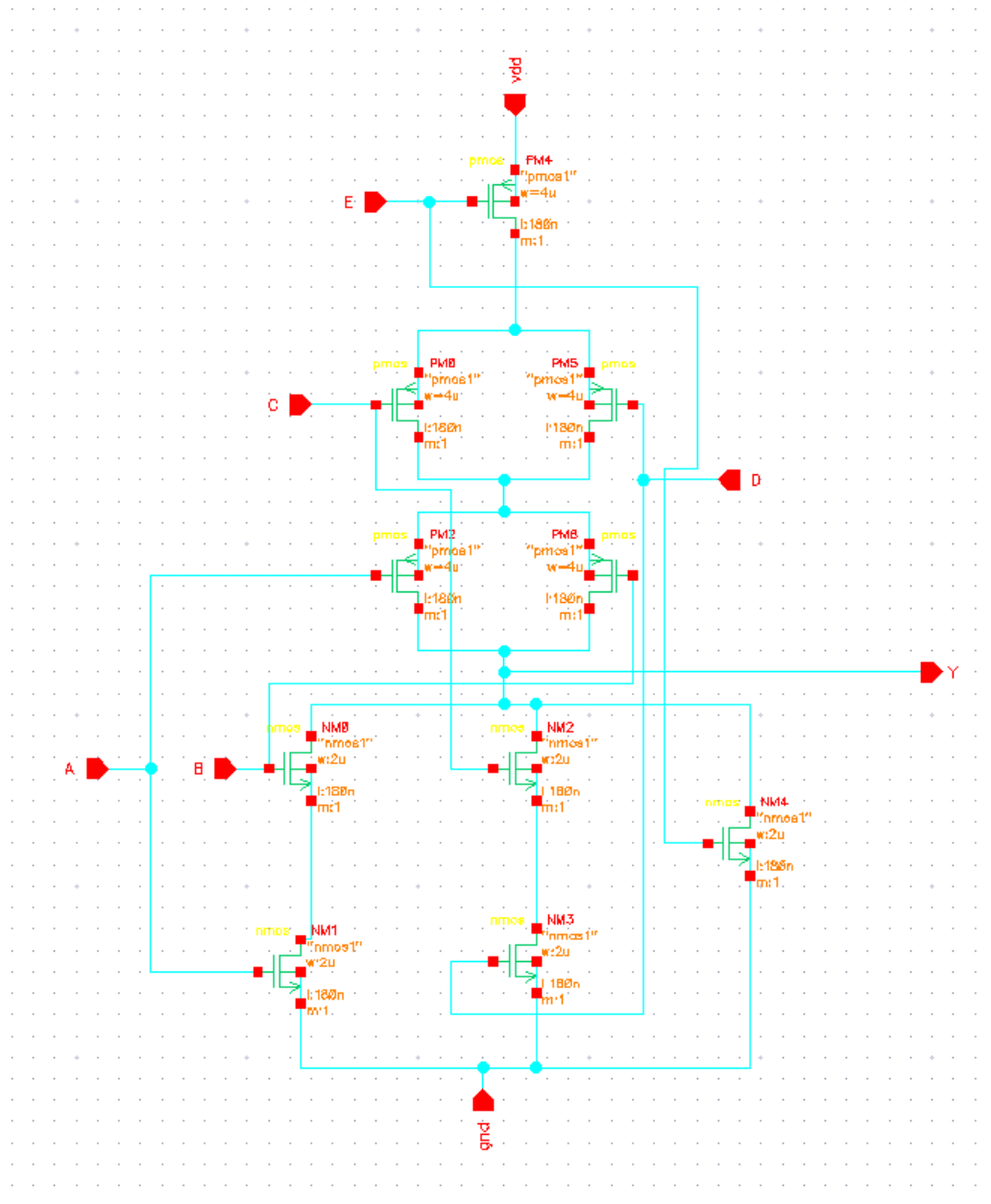
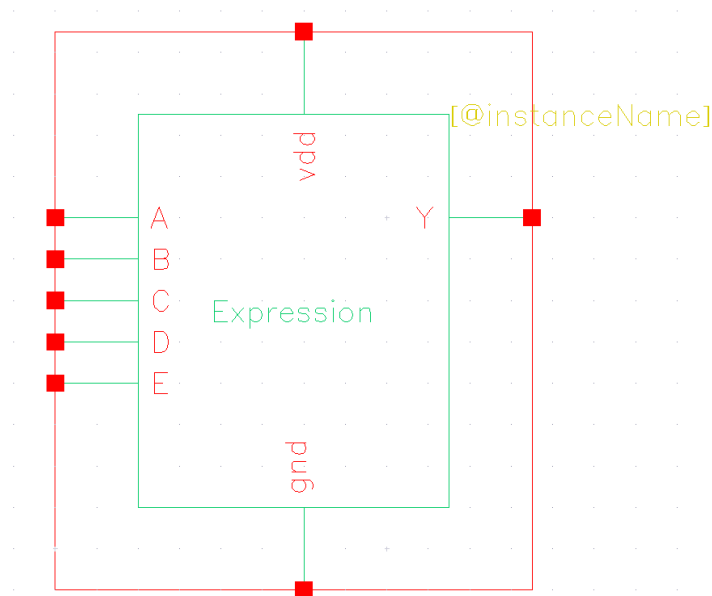


12a. Construct the schematic of the Boolean Expression $Y = \overline{(AB + CD + E)}$ using CMOS Logic. Verify the functionality of the expression, and find out the delay t_d for some combination of input vectors. Tabulate the results.



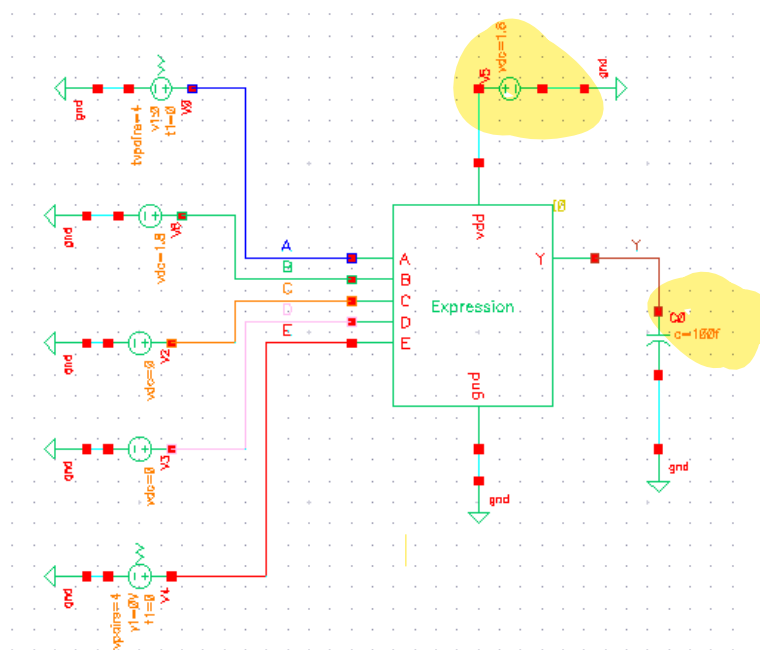
Boolean expression schematic



Boolean expression symbol

Table of components for building the schematic:

Library Name	Cell Name	Properties
gpd180	pmos	W = 4u, L = 180n
gpd180	nmos	W = 2u, L = 180n



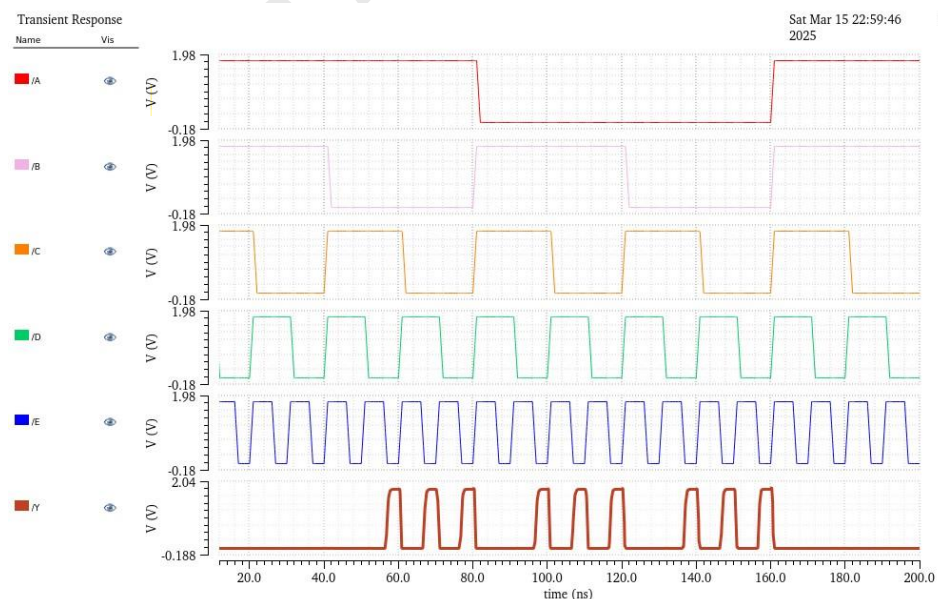
Boolean expression test

Table of components for building the test schematic:

Library Name	Cell Name	Properties
analogLib	Vpulse	For input A , V1 = 0, V2 = 1.8, Period = 160n, Rise time = 1n, Fall time = 1n, Pulse width = 80n
	Vpulse	For input B , V1 = 0, V2 = 1.8, Period = 80n, Rise time = 1n, Fall time = 1n, Pulse width = 40n
	Vpulse	For input C , V1 = 0, V2 = 1.8, Period = 40n, Rise time = 1n, Fall time = 1n, Pulse width = 20n
	Vpulse	For input D , V1 = 0, V2 = 1.8, Period = 20n, Rise time = 1n, Fall time = 1n, Pulse width = 10n
	Vpulse	For input E , V1 = 0, V2 = 1.8, Period = 10n, Rise time = 1n, Fall time = 1n, Pulse width = 5n
analogLib	Vdc	Vdc = 1.8
analogLib	gnd	
analogLib	cap	0.1pF

Table of values to setup for different analysis:

Analysis Name	Settings	Properties
Transient	trans	Stop time = 200ns, moderate



Boolean expression output waveform

12b.Calculation of delay for the condition when A=1, B=1, C=0, D=0, E=0 and note down the waveforms

12c. Calculation of delay for the condition when A=0, B=0, C=0, D=0, E=1 and note down the waveforms